



Adyapan School

Metallurgical Engineering



Duration - 2 months

**Industry
Certification**



Skill India Certified

**250+
Partner Companies**

From extraction to innovation

Explore metallurgy.

From materials to manufacturing excellence - become industry-ready.

This immersive Metallurgical Engineering program takes you beyond foundational concepts into real-world materials processing and analysis. Learn phase diagrams, heat treatment, extractive metallurgy, and material characterization through hands-on case studies and guided applications. With a strong focus on industrial practices and performance optimization, you'll graduate with the ability to analyze material behavior, improve processing techniques, and contribute effectively to modern manufacturing and materials engineering industries with confidence.

8

MODULES

30+

PROGRAM OFFERINGS

20,000+

STUDENTS

250+ PARTNERED COMPANIES



ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course — a platform to launch
your career.

Adyapan Schools was built with a single conviction:
learning works best when it happens in the real world.
We partner with top companies, mentors, and industry
platforms to ensure every student graduates with a
portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live
client projects, giving you the skills and proof-of-work
that employers actually want.

MISSION

To equip ambitious learners with
practitioner-level digital
marketing skills through mentor-
led, project-based education that
bridges the gap between learning
and earning.



VISION

To be India's most trusted
launchpad for the next generation
of marketing leaders — defined
not by degrees but by the real
work.



Everything you need to grow fast

PROGRAM HIGHLIGHTS

Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.

1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.



AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.



Dual Certification

Earn both a Course Completion and Internship Certificate – accredited by Skill India Digital Hub and NSDC.



Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.



Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

8 weeks. 8 modules. Infinite impact.

WEEK 1

Metals, Minerals & the Extraction Framework

- Discussion of Curriculum
- Classification of metals & industrial importance
- Types & geological occurrence of ore minerals
- Routes of metal extraction — pyro / hydro / electro
- Mineral beneficiation: unit-step overview
- Evaluation criteria for beneficiation operations



WEEK 2

Physical Separation of Minerals

- Crushing & grinding- liberation theory
- Screening & classification techniques
- Solid-liquid separation methods
- Automated sorting (sensor-based)
- Energy & efficiency analysis



WEEK 3

Physical Concentration of Minerals

- Gravity separation- jigs, spirals, tables
- Magnetic separation- low & high intensity
- Electrostatic separation
- Froth flotation- theory & reagents
- Flotation circuit design



8 weeks. 8 modules. Infinite impact.

WEEK 4

Fuels, Furnaces & Agglomeration

- Classification & characterisation of fuels
- Coke-making; adiabatic flame temperature calc.
- Types of metallurgical furnaces
- Fuel utilisation & heat exchangers
- Furnace refractories; sintering & pelletisation



WEEK 5

Roasting, Calcination & Reduction Smelting

- Calcination — theory & applications
- Roasting types: oxidising, sulfating, chloridising
- Predominant Area Diagram (PDA) — construction & use
- Smelting thermodynamics — Ellingham diagram
- Applications: Fe, Zn, Pb, Sn, Mg extraction



WEEK 6

Sulfide Extraction & Halide Metallurgy

- Matte smelting theory & matte grades
- Peirce-Smith converting; continuous conversion
- Flash smelting (Outotec process)
- Kroll process- Ti extraction from TiCl_4
- Refining of Cu, Pb, U, Si; new extraction methods



8 weeks. 8 modules. Infinite impact.

WEEK 7

Hydrometallurgy - Leaching to Recovery

- Leaching — acid, alkaline, bacterial; kinetics
- Solvent extraction — extractants & strip circuits
- Ion exchange resins in metal recovery
- Adsorption on activated carbon (CIP/CIL)
- Reduction of ions in solution; application: Au, Ni, Al_2O_3 (Bayer)



WEEK 8

Electrometallurgy & Sustainable Extraction

- Electrochemical cell thermodynamics & Faraday's law
- Electrolytic reduction - Hall-Héroult (Al), Zn EW
- Electrolytic refining - Cu, Ni anode slimes
- Environmental impacts: tailings, SO_2 , GHG, red mud
- Sustainable practices - waste valorisation, circular metal economy



WHO THIS IS FOR

This course is perfect for

Students & Career Switchers

Aspiring Metallurgical & Materials Engineers

Production & Materials Engineering Students

Professionals in Steel, Mining & Manufacturing Industries

Quality Control & Process Engineers

Researchers & Enthusiasts in Materials Processing

CERTIFICATIONS



ALUMNI NETWORK

Our alumni work at world-class companies

Amazon

Adobe

Google

Autodesk

Microsoft

Deloitte

Your career switch is one click away.

Ready to begin? Apply at adyapanschool.com or email us at support@adyapan.com

Apply Now